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Mobile Computing

3/11/26

① Given

Area = 1200 km²

Radius = 1.8 km.

Spectrum = 48 mHz & 200 kHz.

Reuse N = 7 & N = 4

15% Reduction.

Area of Hexagonal cellular

$$A = 2.589 R^2$$

$$A = 2.589 (1.8)^2$$

$$= 8.388$$

No. of Spectrum

$$\Rightarrow \frac{48 \text{ mHz}}{200 \text{ kHz}} = \frac{48000 \text{ Hz}}{200 \text{ Hz}}$$

$$= 240$$

Effictiveness 15%

$$= 240 \times 0.85$$

$$= 204$$

Channels.	N = 7	N = 4
Theoretical Pes cell	240/7 = 34.3	240/4 = 60
Effective Pes cell	204/7 = 29.1	204/4 = 51

Co-Channels

$$D = R\sqrt{3N}$$

$N = 7$

$$D = 1.8\sqrt{3 \times 7} = 1.8\sqrt{21} = 8.249$$

$N = 4$

$$D = 1.8\sqrt{3 \times 4} = 1.8\sqrt{12} = 6.235$$

Reuse

$$Q = \frac{D}{R} = \frac{R\sqrt{3N}}{R} = \sqrt{3N}$$

for $N = 7$

$$\Rightarrow \sqrt{3 \times 7} = \sqrt{21}$$

$$= 4.583$$

for $N = 4$

$$\Rightarrow \sqrt{3 \times 4} = \sqrt{12} = 3.464$$

Minimum reuse ratio ($Q = 4.5$)

$N = 7$ is satisfy the reuse ratio

$N = 4$ is not satisfy the reuse ratio.

(2) Given

recovery = 400 ms.

PLMNs = 40 mins

Cell = 15 at 90 km/h

120ms for BSS

80 ms for NSS

Data Delay.

120+80

= 200

Cells ⇒ 15(200)

= 3000ms.

Normal Delay = 3 seconds.

Recovery Delay

3(400) = 1200ms.

⇒ Recovery Delay = 1.2 seconds.

for Roaming-related Delay.

350+250 = 600

⇒ 2(600) = 1200ms

Roaming-related Delay = 1.2 seconds.

Total Delays:

3s + 1.2s + 1.2s = 5.4 seconds

Dat loss

Dataloss = Rate x Time.

= 512 x 5.4seconds.

= 2764.80kb

= 2.765mb

=>

Normal Delay interrupts the 3seconds.

* Total Delay is 5.4 seconds

* potential Dataloss at 512 kbps is 2.765mb

(3)

i) Given

Base station = 12 Base

Sectors = 3

Transceivers (TRXs) = 4.

TDMA = 8 timeslots & 2 time slots.

Average = 1.2 calls per hour.

Time = 90 Seconds

(4) Given

Network = 240

Size = 18mb

Theoretical Data rate = 20kbps

⇒ Time Slots

240 x 18

= 4320 mb

⇒ GPRS

4 x 20 = 80 kbps

Effictiveness

⇒ 0.85 x 0.82 x 80

= 55.76 Kbps

Total transmitted data

⇒ 4320 x 1.12 = 4838.4 mb

Convest into kilobits

= 4838.4 x 8 x 1000

= 38,707,200 kb

GPRS transfes time

$T = \frac{\text{Data}}{\text{throughput}}$

$$T = \frac{38,707,200}{55.76} = 694175 \text{ seconds}$$

Interruption time :

$$8 \times 2.5 = 20 \text{ seconds}$$

Total

$$694175 + 20 = 694195 \text{ Seconds}$$

Convert into hours

$$\Rightarrow 192.83 \text{ hours.}$$

=> UMTS transfer time

$$T = \frac{38,707,200}{384}$$

$$T = 100800 \text{ seconds}$$

=> Percentage reduction,

$$\Rightarrow \frac{694195 - 100800}{694195} \times 100$$

$$= \boxed{85.48\%}$$

* Deadline = 12 hours

* GPRS take 192.83 hours.

* UMTS take 28 hours

(5) Given.

Telecom = 120GB

Gsm = 2000 mobile

Gsm = 9.6 kpbs

GPRS = 64 kbs

GPRS 10% additional

1mB = 8 megabits.

1GB = 1000mb

Data Per Device.

$= \frac{120GB}{2000} = 0.06GB$

$= 60mB$

Convert into bites

$\Rightarrow 60 \times 8 = 480mb$

GSM Transmission time

9.6 kbps = 0.00096 mbps

$T = \frac{480}{0.0096}$

$= 50000$

GRPS Transmission time.

GIRPS = 13.89 hours

$\Rightarrow 480 \times 1.10$

$= 528mb$

$$T = \frac{528}{0.064}$$

$$= 8250 \text{ Seconds}$$

Convert into hours.

$$= 2.29 \text{ hours}$$

⇒ 5G transfer time

$$25 \times 0.96 = 24 \text{ mbps}$$

Data ⇒ $480 \times 1.04 = 499.2$ mb

Individual $T = \frac{499.2}{24} = 20.8$ Seconds.

Total for 2000 ⇒ $2000 \times 0.40 = 800$.

No. of batches $= \frac{2000}{800} = 2.5$

Total $= 3 \times 20.8 = 62.4$ Seconds.

Percentage reduction with apps

$$\Rightarrow \frac{8250 - 62.4}{8250} \times 100$$

$$= 99.24\%$$